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NATURAL LANGUAGE PROCESSING 2

WEAKLY-SUPERVISED SENTIMENT ANALYSIS ON MOVIE REVIEWS TEACHER–STUDENT DISTILLATION WITH CONFIDENCE-AWARE PSEUDO-LABELING

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NATURAL LANGUAGE PROCESSING 2

CONTEXT

- Weak supervision
- Teacher–student distillation
- TMDb movie reviews

MOTIVATION & RESEARCH QUESTION

- Modern NLP models rely on large labeled datasets
- Manual annotation is:
- expensive
- time-consuming
- prone to subjectivity
- Many real-world datasets are unlabeled

Most NLP benchmarks rely on clean human labels, but in practice, this is rarely available.
So my core research question is : **Can we train a sentiment model using only automatically generated labels, and how reliable would that be?**

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TASK DEFINITION

- Binary Sentiment Classification
  - Positive (POS)
  - Negative (NEG)

Data Source

- Movie reviews collected from the TMDb API

Key Challenges

- Long document-level reviews
- Informal and noisy language
- No ground-truth labels
- Strong class imbalance

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DATASET OVERVIEW

```
================================================================================
[CHECKPOINT] raw_tmdb
shape: (706, 5)
avg char length: 1271.7648725212464
   movie_id                     title      author  \
0       278  The Shawshank Redemption  elshaarawy
1       278  The Shawshank Redemption  John Chard

                                          content                created_at
0               very good movie 9.5/10 29-04-2016  محمد الشعراوىT18:08:41.892Z
1  Some birds aren't meant to be caged.\r\n\r\nTh...  2016-07-10T00:16:50.561Z
================================================================================
Saved CSV: /content/drive/MyDrive/NLP/outputs/raw_reviews_tmdb.csv
```

DATASET(RAW DATA)

- SOURCE: TMDB API
- 706 RAW REVIEWS
- AVG LENGTH ≈ 1271 CHARACTERS

DATA CLEANING & LANGUAGE FILTERING

- ENGLISH-ONLY FILTERING
- HTML & NOISE REMOVAL
- 668 REVIEWS REMAIN
- AVG LENGTH ≈ 1300 CHARACTERS

```
================================================================================
[CHECKPOINT] after_clean_lang
shape: (667, 7)
avg char length: 1336.2608695652175
   movie_id                     title      author  \
1       278  The Shawshank Redemption  John Chard
3       278  The Shawshank Redemption   thommo_nz

                                          content  \
1  Some birds aren't meant to be caged.\r\n\r\nTh...
3  There is a reason why this movie is at the top...

                 created_at  \
1  2016-07-10T00:16:50.561Z
3  2018-05-01T05:51:13.756Z

                                            text  is_en
1  Some birds aren't meant to be caged. The Shaws...   True
3  There is a reason why this movie is at the top...   True
================================================================================
Saved CSV: /content/drive/MyDrive/NLP/outputs/cleaned_reviews_english.csv
```

WHY WEAK SUPERVISION?

- No labels → pseudo-labels
- Labels are noisy by definition
- Reliability must be analyzed

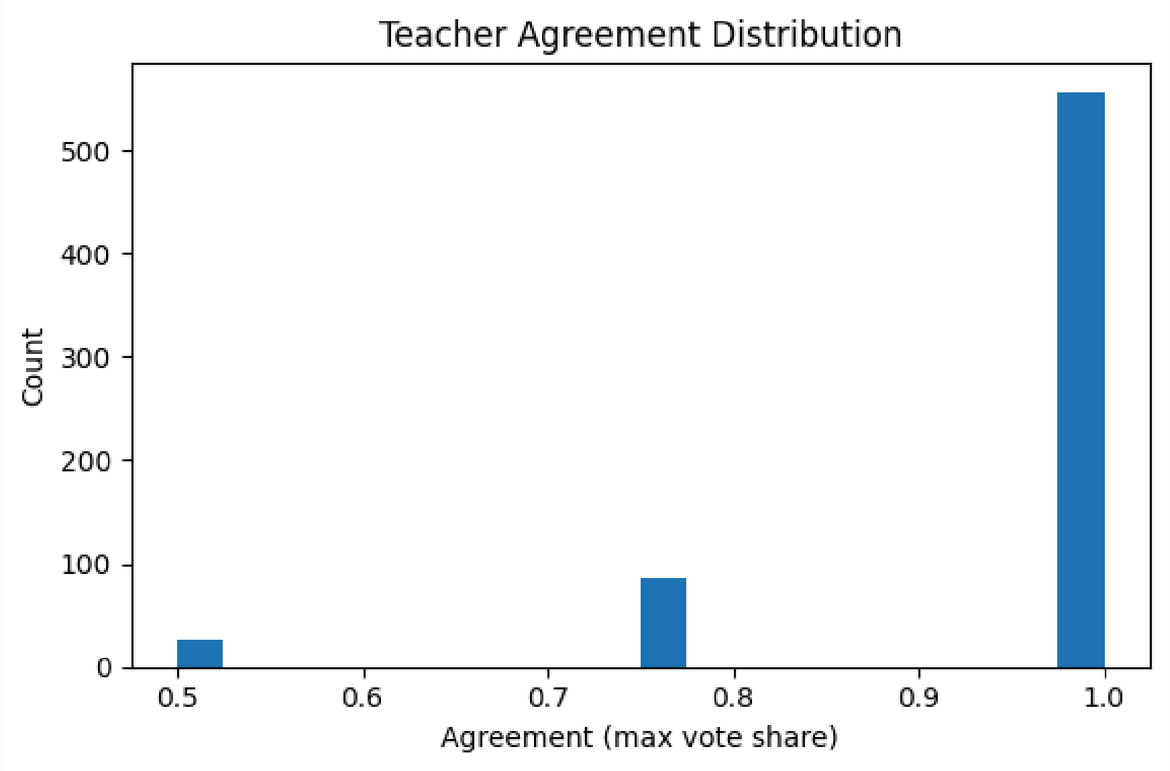
TEACHER ENSEMBLE

- SieBERT
- RoBERTa (SST-2)
- DeBERTa (SST-2)
- DistilBERT (SST-2)

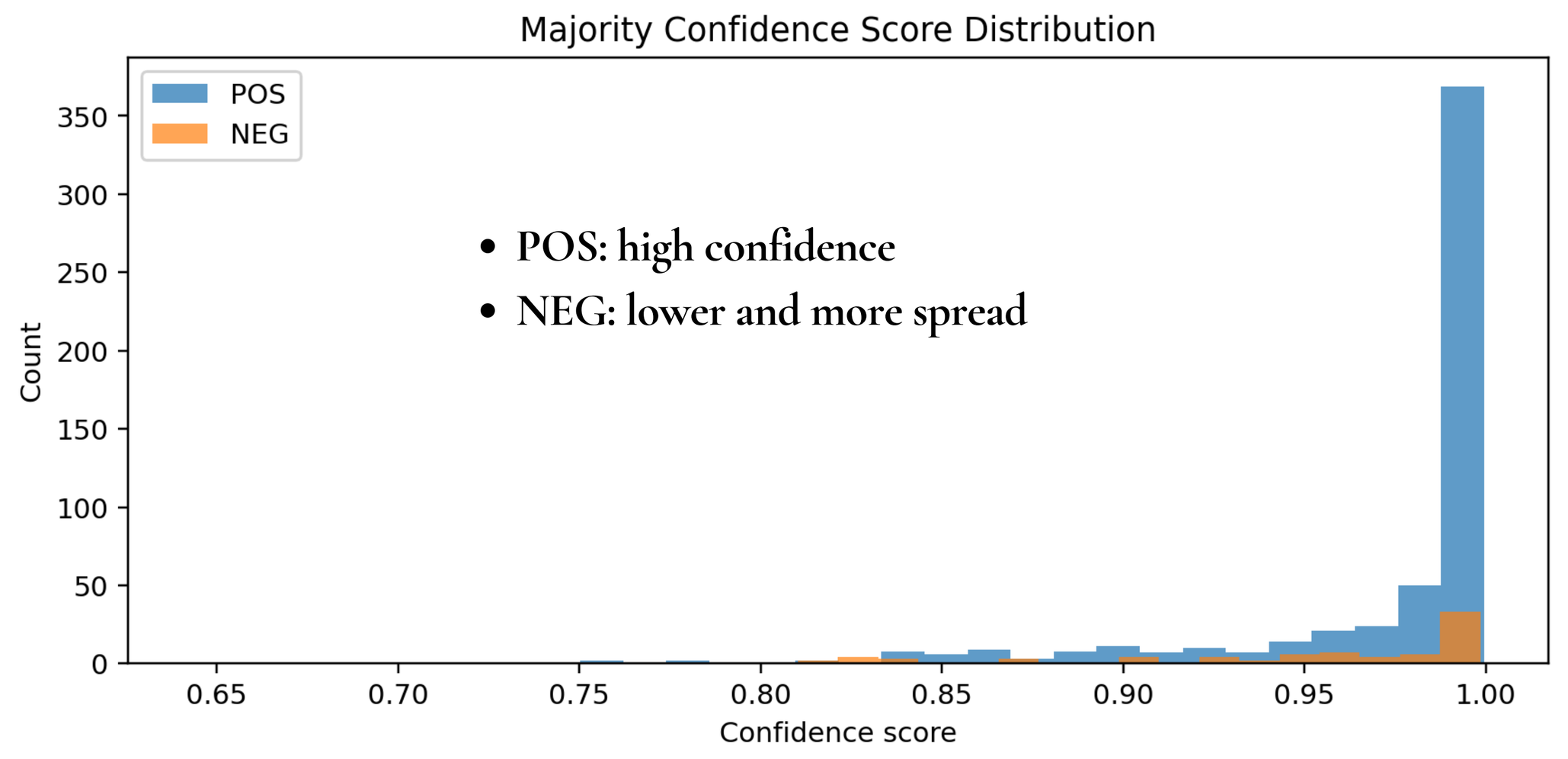
Goal: Reduce individual model bias

MAJORITY VOTING

- 4 teachers → majority voting
- 2–2 splits discarded



CONFIDENCE DISTRIBUTION



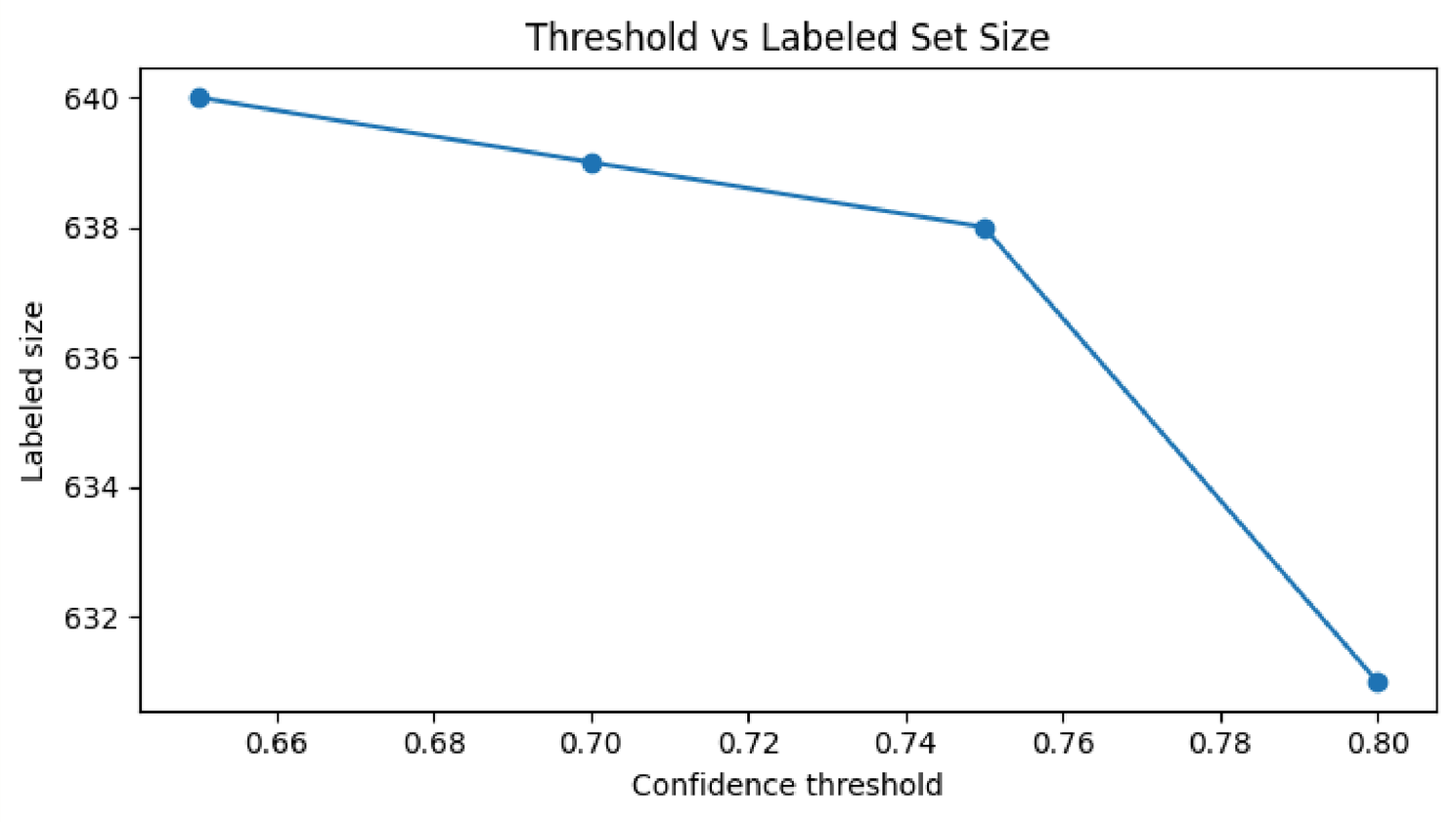
- POS: high confidence
- NEG: lower and more spread

CONFIDENCE THRESHOLDING

- Confidence = mean of agreeing teachers
- Threshold **τ** controls quality vs size

	threshold	size	POS	NEG
0	0.65	639	556	83
1	0.70	638	556	82
2	0.75	637	555	82
3	0.80	630	549	81

THRESHOLD ANALYSIS



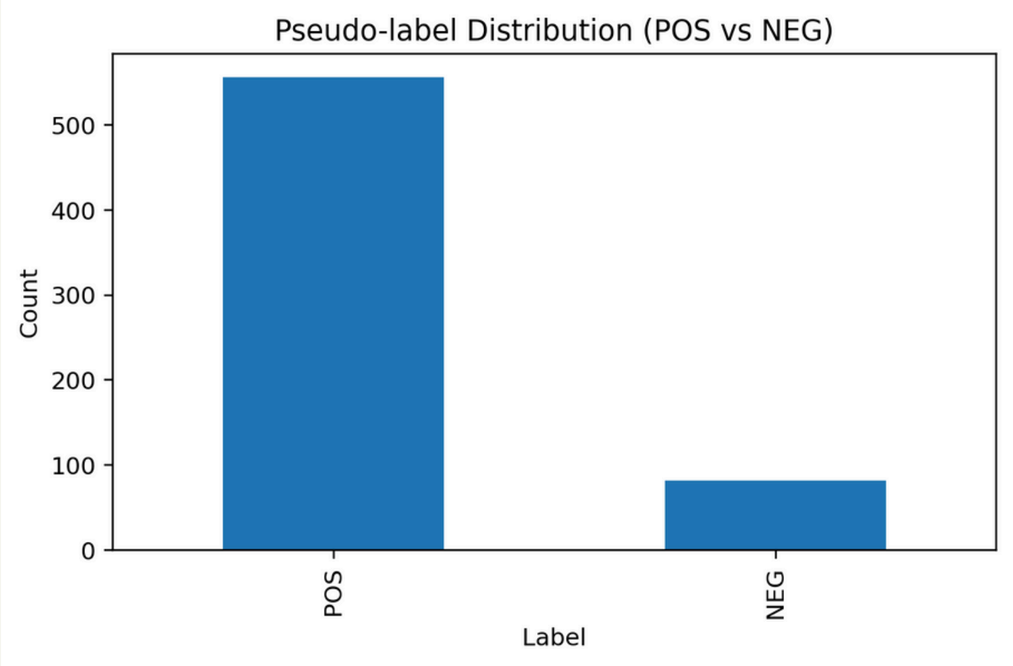
$$c(x) = \frac{1}{|\mathcal{A}|} \sum_{t \in \mathcal{A}} p_t(\hat{y} \mid x).$$

A sample is kept if:

- ≥ 3 teachers agree
- Mean confidence ≥ **τ**

The threshold **τ** directly controls a quality–quantity trade-off. As **τ** increases, the labeled dataset size decreases, but the average label reliability increases. In our experiments, **τ**=0.75 provided the best balance between dataset size (≈638 samples) and label confidence.

Trade-off between label quality and dataset size



PSEUDO-LABELING

Pseudo-Labels

- Generated by pretrained models
- Replace human annotations
- Noisy by nature
- POS: 556
- NEG:82
- Several class imbalance

METHODOLOGY OVERVIEW

```
==============================================================================
TABLE 1 (Dataset Statistics) — I'll just copy this into my LaTeX report later
            #Samples  Positive (%)  Negative (%)
Train          459.0          87.1          12.9
Validation      82.0          86.6          13.4
Test            96.0          87.5          12.5
==============================================================================
```

- Stratified split
- Same imbalance across splits

BASELINE MODELS - RESULT

- TF–IDF + Logistic Regression
- TF–IDF + Linear SVM

Why baselines?

- Robust to label noise
- Strong reference point
- Interpretable behavior

	model	ACC	BACC	MacroF1
0	TFIDF+LogReg	0.895833	0.583333	0.614767
1	TFIDF+LinearSVM	0.875000	0.500000	0.466667

Baseline models achieve high accuracy but substantially lower Macro-F1 scores, indicating poor performance on the minority negative class.

Macro-F1 ≈ 0.61 (best baseline achieved)

• Initial expectation: student model would outperform baselines

STUDENT MODEL - RESULT

- DistilBERT fine-tuned on pseudo-labeled data

Why DistilBERT?

- Lightweight and computationally efficient
- Lower risk of overfitting on noisy labels
- Suitable for small and weakly supervised datasets
- • Model capacity was intentionally limited

```
DistilBERT -> ACC: 0.8958333333333334 BACC: 0.6904761904761905 MacroF1: 0.7209302325581395
```

Because over 85% of the dataset is positive, predicting the majority class alone already yields high accuracy. Therefore, accuracy does not reflect the model's ability to detect negative sentiment.

	precision	recall	f1-score	support
NEG	0.411765	0.583333	0.482759	12.00000
POS	0.936709	0.880952	0.907975	84.00000
accuracy	0.843750	0.843750	0.843750	0.84375
macro avg	0.674237	0.732143	0.695367	96.00000
weighted avg	0.871091	0.843750	0.854823	96.00000

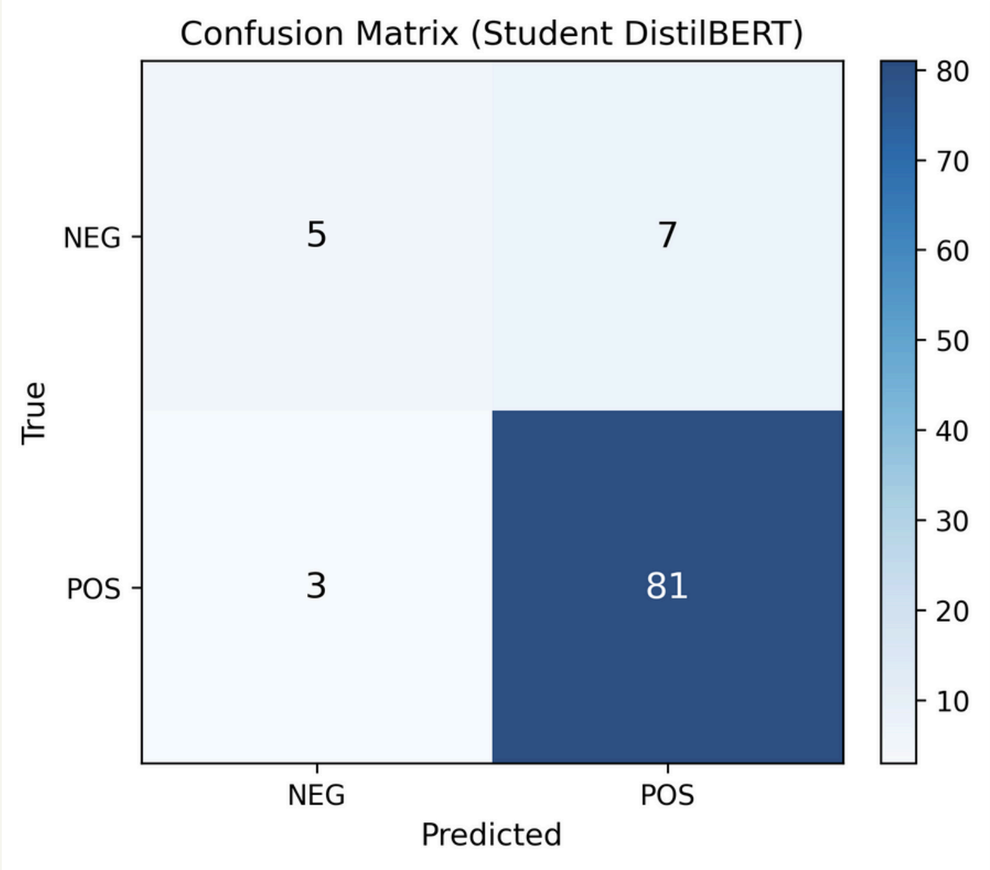
Class-wise evaluation shows strong performance on positive reviews but reduced recall and F1-score for negative sentiment.

Why Accuracy Is Misleading

- Dataset is heavily imbalanced
- Majority class dominates predictions
- High accuracy does not imply balanced performance
- Unexpectedly low recall for the negative class

Macro-F1 is the primary evaluation metric.

The confusion matrix illustrates that most errors occur when negative reviews are misclassified as positive, despite high overall accuracy.



Confusion Matrix Analysis

- Negative reviews are frequently predicted as Positive
- Strong bias toward the majority class

More than half of the negative samples are misclassified as positive, confirming that minority-class recall is the primary weakness of the system.

ERROR ANALYSIS

- Errors concentrated in the Negative class
- Negative sentiment is harder to detect
- Bias propagates from teachers to the student

```
Confusion Matrix (Table Form):
          Pred_NEG  Pred_POS
True_NEG         5         7
True_POS         3        81
Saved table: /content/drive/MyDrive/NLP/outputs/tables/confusion_matrix_student_table.csv
              precision    recall  f1-score    support
NEG            0.625000  0.416667  0.500000  12.000000
POS            0.920455  0.964286  0.941860  84.000000
accuracy       0.895833  0.895833  0.895833   0.895833
macro avg      0.772727  0.690476  0.720930  96.000000
weighted avg   0.883523  0.895833  0.886628  96.000000
Saved table: /content/drive/MyDrive/NLP/outputs/tables/student_classwise_report.csv
```

Numerical confusion matrix confirms that minority-class errors dominate overall performance degradation.

The observed errors are not primarily caused by model capacity, but by systematic bias in the pseudo-labels generated by the teacher ensemble.

QUALITATIVE ANALYSIS

- Positive reviews emphasize emotional and evaluative language
- Negative reviews focus on criticism and dissatisfaction

WordClouds demonstrate that pseudo-labels capture semantically meaningful sentiment-related vocabulary.

WHAT DID NOT WORK

- LSTM models
- Oversampling
- Class-weighted loss

KEY INSIGHT

In weak supervision:
Label quality matters more than model complexity.

- Strong baselines remain competitive
- Larger models cannot fix noisy labels
- Bias analysis is essential

Severe class imbalance explains why accuracy alone is insufficient and motivates the use of Macro-F1.

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CONCLUSION

- Weakly supervised sentiment analysis is feasible
- Ensemble agreement improves label reliability
- Confidence filtering is essential
- Accuracy alone is misleading under imbalance

This study demonstrates that weak supervision is not a shortcut to labeled data, but a controlled trade-off between annotation cost and label reliability.
Final pseudo-labeled dataset after confidence filtering and teacher agreement.

```
================================================================================
[CHECKPOINT] final_labeled_set
shape: (637, 26)
   movie_id                     title      author  \
1       278  The Shawshank Redemption  John Chard
3       278  The Shawshank Redemption   thommo_nz

                                            content  \
1  Some birds aren't meant to be caged.\r\n\r\nTh...
3  There is a reason why this movie is at the top...

                 created_at  \
1  2016-07-10T00:16:50.561Z
3  2018-05-01T05:51:13.756Z

                                              text  is_en    t1_raw  \
1  Some birds aren't meant to be caged. The Shaws...   True  POSITIVE
3  There is a reason why this movie is at the top...   True  POSITIVE

   t1_score t1_lab  ...   t4_raw  t4_score t4_lab t_votes_pos  t_votes_neg  \
1  0.997052    POS  ...  LABEL_1  0.849160    POS           4            0
3  0.998910    POS  ...  LABEL_1  0.999437    POS           4            0

  t_major_lab t_major_score  vote_strength agreement  label
1         POS      0.847546            4-0       1.0    POS
3         POS      0.999303            4-0       1.0    POS

[2 rows x 26 columns]
================================================================================
label
POS    555
NEG     82
Name: count, dtype: int64
Saved CSV: /content/drive/MyDrive/NLP/outputs/pseudo_labels_final.csv
```

FUTURE WORK

- Small human-labeled validation set
- Confidence calibration
- Noise-aware loss functions
- Curriculum or self-training strategies
- Limited by time and computational resources

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THANK YOU...